chapter, and usually involves some form of price prediction. The risk model in turn ensures checks and balances against adverse positions and exposures are in place. The transaction cost model aids the portfolio construction model to decipher what trades and/or positions are necessary to achieve the trading goal, passing these decisions onto the execution model to enact.

Transient to the core of the black box is accurate and well-curated data. High frequency, or tick-by-tick, data is vast. The number of observations for a single day in a liquid market is around 30 years of daily data (Dacorogna et al., 2001). Rigorous research on accurate data is at the very core of a successful high frequency strategy. Prior to detailing some strategies and an evaluation framework, we shall take a closer look at high frequency data.

DATA AND RESEARCH

High frequency data originates from the financial markets. By its very nature it is irregularly spaced in time, however, and with the sheer volume being reported by liquid markets can only be understood using continuous dynamics (Hanif and Protopapas, 2013). Financial data providers usually report hundreds of thousands of prices for a single market a day. Dacorogna et al. (2001) argue, correctly, that high frequency data should be the primary concern of those interested in understanding the financial markets, especially given the effect of market dynamics on everyday investors. Unfortunately this is not the case with most academic and empirical financial literature for, they argue, two reasons.

Firstly, it is costly and resource-intensive to collect, store, manipulate and curate high frequency data. This is precisely why most available data is either daily or lower frequencies. The second reason is more subtle. Most statistical modeling and machine learning tools assume time-series homogeneity; that is they assume regularly spaced data points. Little work has been done to look into irregular data with Hanif and Protopapas (2013), described above, working towards bridging the gap between regularized and irregular time series. Homogeneous financial time-series do not come from the markets but are the result of post-processing on raw tick-by-tick data.

Time-series operators such as interpolation methods are used to transform inhomogeneous time-series into a homogeneous time-series ready for analysis. Variables which are commonly used to capture intraday dynamics include price, return, realized volatility, bid-ask spread, tick frequency,